

April 25, 2018

To the Editors:

I would like to submit an original article entitled “Estimating Population Mean Power Under Conditions of Heterogeneity and Selection for Significance” by Jerry Brunner and Ulrich Schimmack, for possible publication in *Meta-Psychology*. The paper develops and compares four methods for estimating the mean power of a diverse collection of statistical tests. Note that this is not statistical power assessed for *a priori* values like Cohen's “small,” “medium” and “large” effect sizes, but power at the true effect sizes in the studies.

The paper has a theoretical part and a simulation part. The theoretical part is based on the following idea. The suppression of non-significant results (publication bias) creates two populations of power values, the population before selection, and the sub-population for which the results happened to be significant. We have discovered a set of fundamental and surprisingly simple principles connecting these two populations of power. For example, the increase in population mean power due to selection for significance equals the population variance of power before selection divided by the population mean of power before selection. We believe these principles to be previously unknown. Proofs are given in the appendix.

The simulations are notable in several respects.

- They are very large-scale. Our typical simulation study is based on a multi-way factorial design, with 10,000 simulated meta-analyses in each treatment combination.
- We investigate a wider range of tests than usual -- F and chi-squared with varying degrees of freedom, not just z or t as in the typical simulation study of effect size.
- The simulations, which would have taken decades to run on a single processor, wrote results to plain text data files that were analysed in a separate step to produce the tables in the paper. These raw data files are available for further analysis, as part of the supplementary materials.
- Using the principles in the theoretical section, we were able to adjust the distributions of sample size and effect before selection so that population mean power after selection (the quantity we were trying to estimate) came out to nice round numbers like 0.25, 0.50 and 0.75. This is very difficult to accomplish by trial and error, and makes the results a lot more readable.

The paper is typeset in LaTeX, using bibtex and the apa6 document class to produce references in APA format. As requested, an editable version of the paper is included with the submission materials. To make it actually useable, it is provided as a zip archive that includes the graphics files and the .bib file as well as the .tex document.

The supplementary materials, which are quite extensive, are currently mounted on a University of Toronto server at <http://www.utstat.toronto.edu/~brunner/zcurve2018>. There are links from the pdf version of the paper to the supplementary materials. Of course I have no objection to putting the supplementary materials on your webserver, but the data files written by the simulation code are millions of lines – around 300 MB of plain text. You might want to leave them where they are, at least during the review process. This is just a suggestion.

The Submission Preparation Checklist includes “Account on OSF – linked in cover letter.” I have created an account on OSF, associated with the name Jerry Brunner and the email address brunner@utstat.toronto.edu. Unfortunately, I cannot see how to “link” to it. Please let me know what I can do to address this requirement.

Sincerely,



Jerry Brunner